

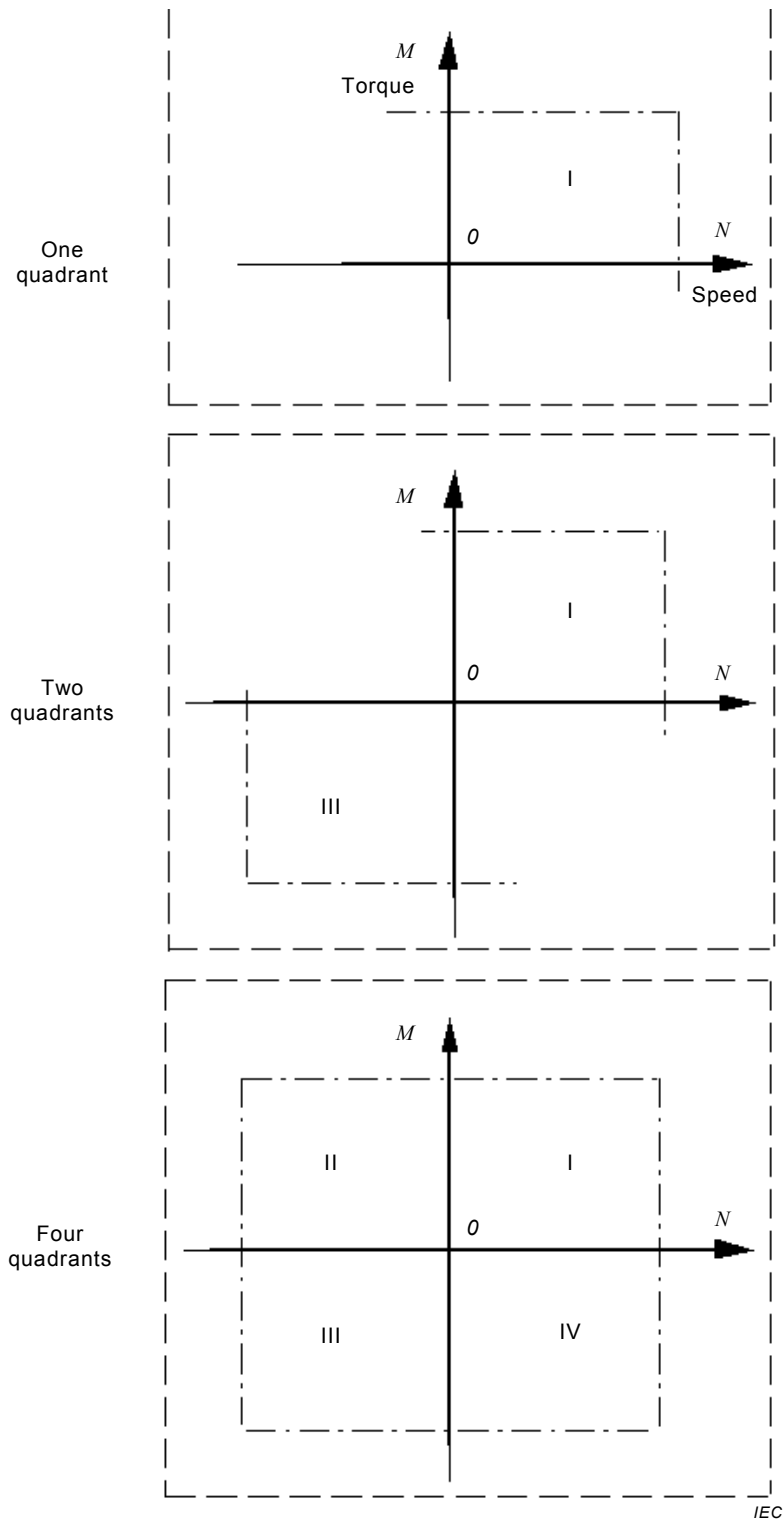


Figure 3 – Operating quadrants

4 Ratings and specifications for the act of installing, commissioning and operation

4.1 General

The requirements in 4.3 to 4.13 provide a list of requirements relevant for the specification of a *BDM/CDM/PDS*. More severe requirements might be specified if considered relevant for the considered application.

A *BDM/CDM/PDS* is typically used for a specific application in a specific environment under specific conditions, in which the product has to operate, be transported or stored. These conditions include, but are not limited to the electrical, electro-magnetic, mechanical, thermal, and chemical environment as well as requirements on the functionality, safety and functional safety. These conditions are known by the *customer* or product standard committees using this document as a reference document and will need to be specified.

The *manufacturer* of the *BDM/CDM/PDS* shall specify which requirements of IEC 61800-2 apply to his equipment.

In order to ensure consistency and avoid conflicting requirements across the IEC 61800 series, some of the subclauses in 4.3 to 4.13 refer directly to other parts of the IEC 61800 series.

4.2 *BDM/CDM/PDS* characteristics and topology

4.2.1 General

Subclauses 4.2.2 and 4.2.3 are intended to be informative regarding common characteristics and topology for *PDS*'s. The information in these clauses shall not be construed as requirements.

4.2.2 *BDM/CDM/PDS* characteristics

A.C. *PDS*'s are a type of electronic power conversion equipment, which provide speed, current or torque control for a.c. electric *motors*. Because a.c. induction *motors* dominate industrial applications, a.c. *PDS*'s designed to operate a.c. induction *motors* are the most numerous. However, many high *efficiency motor* technologies require a *CDM* for operation. Therefore the use of a.c. *PDS*'s with other types of a.c. *motors* will increase. The use of *CDM* with permanent magnet (PM) *motors* has already reached a significant level.

Some significant characteristics of low voltage *BDM/CDM/PDS*'s include the following.

- *BDM/CDM/PDS*'s are commonly available with *output power* ranging from 0,2 kW to several thousand kW.
- Most industrial *BDM/CDM/PDS*'s are designed to be powered from a 3 phase a.c. supply.
- Some low power *BDM/CDM/PDS*'s are designed to be powered from single phase a.c.
- Many *BDM/CDM/PDS*'s are designed to receive d.c. power from a *port* connecting d.c. links of more than one *PDS*. Many *PDS*'s may receive power from both an a.c. supply and from a *port* connecting d.c. links
- *BDM/CDM/PDS*'s vary the speed of an a.c. *motor* by controlling the frequency and voltage of the power provided to the *motor*.
- The most common *BDM/CDM/PDS*'s are designed to control 3 phase induction *motors* with voltage ratings such as 240 V, 400 V, 480 V, 600 V, and 690 V.
- Some *BDM/CDM/PDS*'s are designed for use with stepper, or switched reluctance *motors*.
- Many *BDM/CDM/PDS*'s are designed for use with permanent magnet *motors*.
- Energy *efficiency* of the typical a.c. *BDM/CDM* is typically very high. In general it is beneficial to reduce energy lost to heating effects in order to minimize size and operating cost.
- Most a.c.. *PDS*'s return power from the *motor* to the d.c. link during periods when the *motor* operates as a generator (operation in II and IV *quadrants*). (see Figure 3)
- Many a.c. *PDS*'s are provided with a dynamic brake (also called "chopper brake" or "brake chopper") in order to manage power returned from the *motor* to the d.c. link during periods when the *motor* operates as a generator.

- Regenerative *PDS*'s are designed to return power from the *d.c. link* of the *BDM/CDM/PDS*'s to the a.c. supply. In some implementations, power conversion from the *d.c. link* to the a.c. supply may be accomplished in a subsystem separate from the *BDM/CDM/PDS*.
- *BDM/CDM/PDS*'s for use with a.c. induction *motors* are available with different control algorithms which optimize cost, and speed/torque regulation for different applications. Examples include
 - volts/hertz control,
 - sensorless vector control,
 - flux vector control,
 - sensorless flux vector control,
 - field oriented control,
 - sensorless field oriented control.

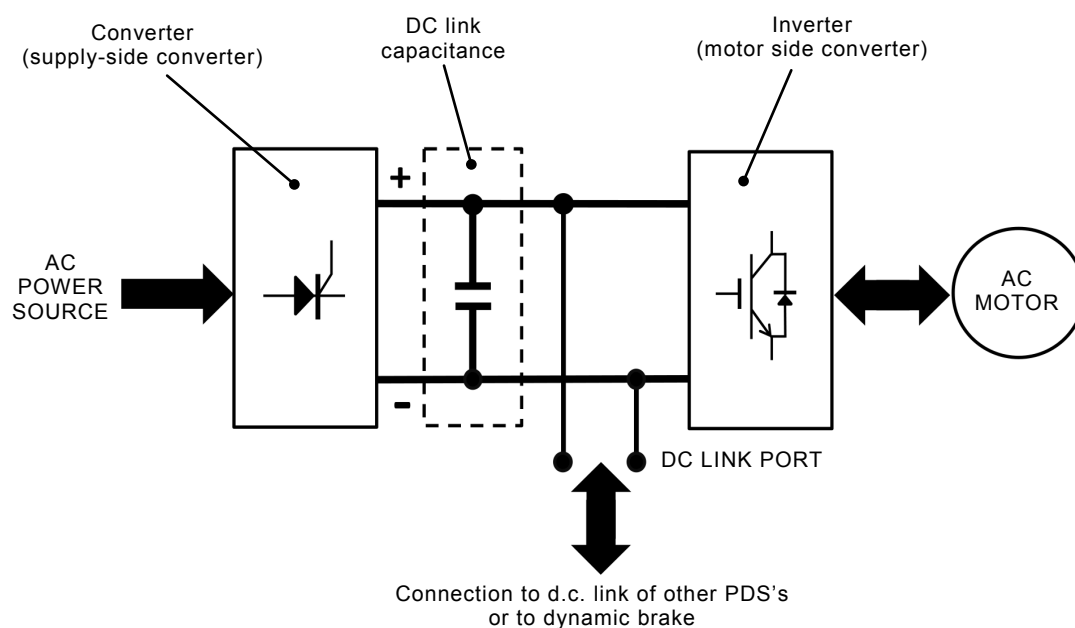
4.2.3 Basic topology for *BDM/CDM/PDS*'s

The most commonly-used topology for low voltage *BDM/CDM/PDS*'s is that of the voltage source *converter* (VSC). In a VSC, a supply-side *converter* changes electricity from the a.c. supply to d.c. Capacitance is used to smooth the d.c. output of the *converter*, and provide short-term energy storage. The d.c. output of the supply-side *converter*, sometimes referred to as the *d.c. link*, supplies energy to the *motor-side converter*, also known as an *inverter*. The *inverter* typically uses PWM (Pulse Width Modulation) to power an a.c. *motor* and permit control of *motor* speed and torque.

Figure 4 below illustrates a common *BDM/CDM/PDS*'s topology. In the *BDM/CDM/PDS*'s of Figure 4, energy flow is unidirectional from the a.c. power source into the *converter*. Energy flow between the *inverter* and *motor* is bidirectional depending upon the dynamics of the mechanical load on the *motor*.

The *d.c. link* port allows exchange of energy with the *d.c. link port* of other *BDM/CDM/PDS*'s or with a dynamic brake. If the *d.c. link port* is connected to the *d.c. link* ports of other *PDS*'s, it is possible to share energy either from the *converter*, or energy developed by the *motor* during operation in *quadrants* II and IV, with other *PDS* operating in *quadrants* I and III. Otherwise, the *d.c. link port* may be connected to an external dynamic brake in order to dissipate excess energy when the voltage of the *d.c. link* exceeds desired limits. It is also possible to connect an external regenerative unit to the *d.c. link* and feed power back to a.c. mains.

The *d.c. link* connection should be designed and protected well. In poorly designed *d.c. link* systems it is possible for a low power *CDM* to feed a high power *CDM*. This may lead to destruction of the low power *CDM*. Also, if no suitable protection of the *d.c. link* connection has been provided, e.g. by fuses, fault conditions can lead to the destruction of one or more units connected to the common *d.c. link*.



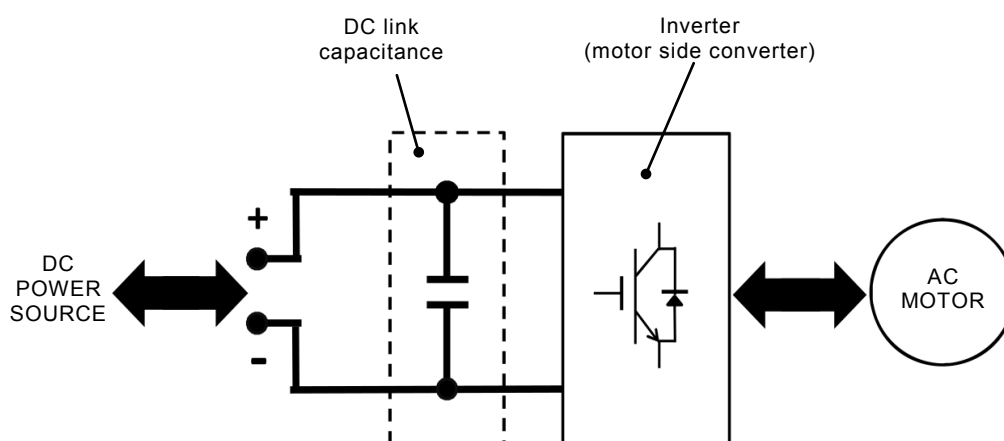
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Figure 4 – Typical **BDM/CDM/PDS**

BDM/CDM/PDS's with common d.c. link ports can be constructed without a supply-side converter. These *PDS*'s are intended for use together with other power conversion units. In these configurations, the *d.c. link port* becomes a means of energy exchange between the different units. Examples of these units include:

- *BDM/CDM/PDS*(s) with a d.c. link port (Figure 5),
- dedicated supply-side converter(s) (no figure),
- dedicated dynamic brake(s) (no figure).

A *PDS* with a d.c. link port is illustrated in Figure 5 below.



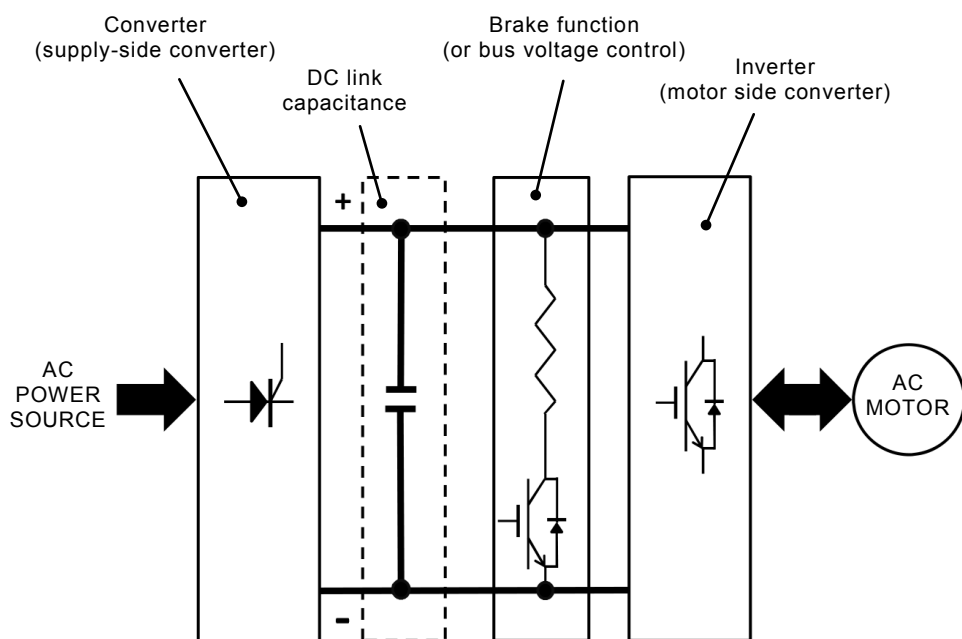
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Figure 5 – Common d.c. link **BDM/CDM/PDS**

When a *BDM/CDM/PDS* operates in *quadrants* II and IV, the inertia of the *motor* and connected load, or sometimes potential energy in the connected load, results in energy being generated by the *motor* and returned through the *inverter* to the d.c. link port. During these periods, the energy returned to the d.c. link port is often managed using one or more of the following options:

- *BDM/CDM/PDS(s)* with d.c. link ports which use(s) energy available from the d.c. link to power other *motors*;
- *dynamic braking* which dissipates excess energy from the d.c. link using resistors (Figure 6);
- return of energy in the d.c. link to an a.c. power system for use by other loads.

Figure 6 illustrates a *BDM/CDM/PDS* which incorporates a dynamic brake. Dynamic brakes are also commonly available as stand-alone units for connection to a d.c. link *port*.

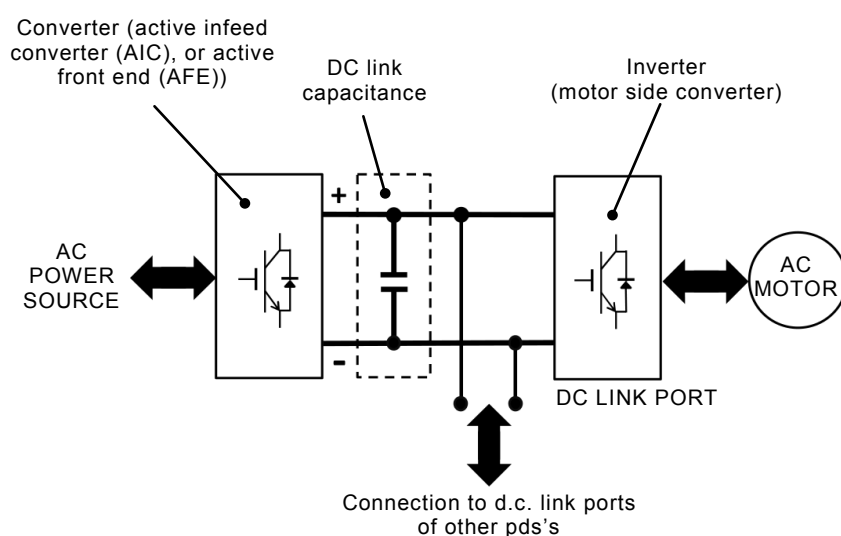


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Figure 6 – *BDM/CDM/PDS* with brake

Figure 7 illustrates a regenerative *PDS* which can return energy to the a.c. supply using a supply side *converter* which also can operate as an *inverter*, also known as *active infeed converters* (AIC's).

Additional information regarding AIC is available in IEC TS 62578.



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Figure 7 – BDM/CDM/PDS with AIC

4.3 Ratings

4.3.1 General

The input and output ratings shall be specified by the *manufacturer* according to the requirements in 4.3.2 to 4.3.6 as shown Table 5.

Table 5 – Overview of input and output ratings of the BDM/CDM/PDS

Input ratings			Output ratings		
BDM	CDM	PDS	BDM	CDM	PDS
Voltage (U_{VN}) [V]	Voltage (U_{LN}) [V]	Voltage (U_{LN}) [V]	Voltage (U_{aN1}) [V]	Voltage (U_{AN1}) [V]	-----
Current (I_{VN}) [A]	Current (I_{LN}) [A]	Current (I_{LN}) [A]	Current (I_{aN}) [A]	Current (I_{AN}) [A]	Torque (M_N) [N·m]
Power (S_{VN} or P_{VN}) [kVA] or [kW]	Power (S_{LN} or P_{LN}) [kVA] or [kW]	Power (S_{LN} or P_{LN}) [kVA] or [kW]	Power (S_{aN} or P_{aN}) [kVA] or [kW]	Power (S_{AN} or P_{AN}) [kVA] or [kW]	Power (P_{SN}) [kW]
Frequency (f_{VN}) [Hz]	Frequency (f_{LN}) [Hz]	Frequency (f_{LN}) [Hz]	Frequency (f_{aN}) [Hz]	Frequency (f_{AN}) [Hz]	Speed (N_N and N_{NMax}) [r/min]

NOTE 1 In case no transformer nor other optional device such a filter are installed between mains input (i.e. CDM/PDS input) and BDM input, the ratings of CDM/PDS input and those of BDM input are same values, i.e. $U_{VN} = U_{LN}$ and so on.

NOTE 2 In case no transformer nor other optional device such a filter are installed between BDM output and motor input (i.e. CDM output), the ratings of CDM output and those of BDM output are same values, i.e. $U_{aN1} = U_{AN1}$ and so on.

NOTE 3 The BDM/CDM output power rating can be provided in either apparent power [kVA] or active power [kW]. Apparent power can be calculated using voltage and current.

NOTE 4 The BDM/CDM/PDS current, voltage and frequency ratings can be provided within a range of values.

4.3.2 Input ratings

4.3.2.1 Input voltages and frequencies

The *BDM/CDM/PDS* input voltage and frequency ratings/range shall be specified by the *manufacturer*.

Preferred values are listed in Table 6:

Table 6 – Standard voltages as specified in IEC 60038

At 50 Hz	At 60 Hz
100	100
110	---
---	115
---	120
200	200
---	208
220	220
230	230
240	240
380	---
400	400
415	---
440	440
	460
---	480
500	---
---	575
660	---
---	600
690	---
NOTE 1 Voltage tolerances are provided in IEC 61800-3.	
NOTE 2 Standard voltages as specified in IEC 60038	

For compliance see 5.4.2.4.1

4.3.2.2 Input current

The *BDM/CDM/PDS* input current rating shall be specified by the *manufacturer*.

The specified *input current* includes the current required by the auxiliaries if they are supplied from the same supply of the *BDM/CDM/PDS*.

These values shall be stated by *the manufacturer*.

For compliance see 5.4.2.4.3.

4.3.3 Output ratings

4.3.3.1 BDM/CDM continuous output ratings

Continuous output ratings shall be stated by the *manufacturer* and shall be in terms of the *motor* output of the *BDM/CDM*:

- fundamental a.c. voltage (U_{aN1}/U_{AN1});
- *rated output current* (I_{aN}/I_{AN}); and
- *output frequency range*;
- *rated maximum apparent output power* (S_{aN}/S_{AN}) [kVA] or *maximum output active power* (P_{aN}/P_{AN}) [kW]

Rating in term of U_{aN1} , U_{AN1} and I_{aN}/I_{AN} allows use of direct measurement techniques and adequately addresses conductor current capacity.

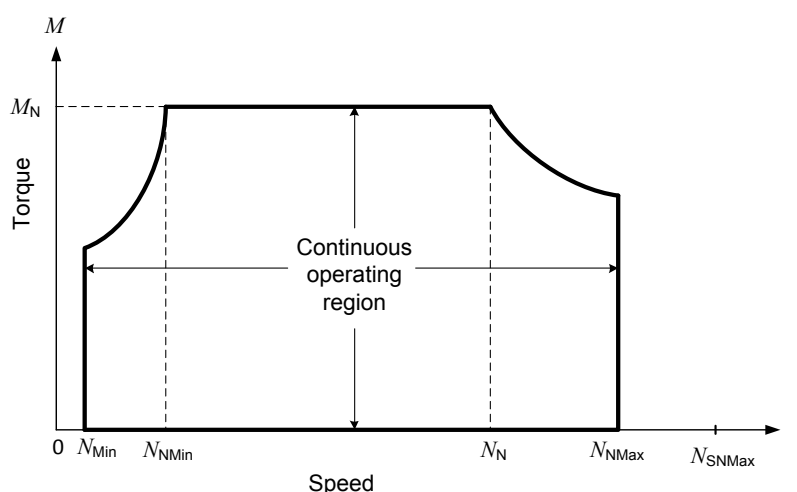
For compliance see 5.4.2.5.2, 5.4.2.5.3, 5.4.2.5.4.

NOTE When *CDM* and *motor* are not provided by the same *manufacturer/supplier*, information can be exchanged to define proper performance and compatibility of *CDM* and *motor*.

4.3.3.2 PDS continuous output ratings

Continuous output ratings shall be stated by the *manufacturer* and shall be in terms of the *motor* shaft of the *PDS*:

- *rated torque* (M_N) [N·m];
- *rated speed* (N_N) [r/min];
- *maximum rated speed* (N_{NMax}) [r/min];
- *minimum rated speed* (N_{NMin}) [r/min];
- *minimum speed* (N_{Min}) [r/min];
- *maximum rated safe speed* (N_{SNMax}) [r/min];
- *rated output power* (P_{SN}) [kW].



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Figure 8 – Example of operating region of a *PDS*

For compliance see 5.4.2.5.2, 5.4.2.5.3, 5.4.2.5.4

4.3.3.3 Overcurrent and torque capability

Besides the rated value under continuous load conditions of *BDM/CDM/PDS*, the *manufacturer* may assign additional values of the rated current, each under specified overload conditions, i.e. a *BDM/CDM/PDS* equipment may be specified by the *manufacturer* with different overload ratings for different types of load. The *output overload capability* applies to the *rated speed range*.

The *output overload capability* of a *BDM/CDM/PDS* can be specified as an intermittent load duty, or as a repetitive load duty. A wider classification, as well as calculation methods, can be found in IEC TR 61800-6.

NOTE The relationship between the overload currents specified and the torque developed by the *motor* is not covered by the above, i.e. *PDS* overload is defined as the overload currents of the *BDM/CDM* and not as the torque of the *motor*.

Special overload conditions can be specified. E.g. overload magnitude and duration may be the subject of such a specification. Examples of typical overload magnitudes and duration are given in 4.10, IEC 60146-1-1 and IEC TR 61800-6.

For any type of duty cycle, the r.m.s value of the current over the complete cycle shall not exceed the rated current. Table 7 and Figure 9 show 6 typical examples of a 1 min overload with a 10 min and 60 min. load cycle.

Table 7 – Example of reduced maximum continuous load as a function of an overload

Overload		Reduced continuous load	
Amplitude I_{aM} [p.u. of rated]	Duration T_{aM} [min]	Maximum amplitude of I_{aR} [p.u. of rated]	Duration T_{aR} [min]
1,5	1	0,928	9
1,5	1	0,989	59
1,25	1	0,968	9
1,25	1	0,995	59
1,1	1	0,988	9
1,1	1	0,998	59

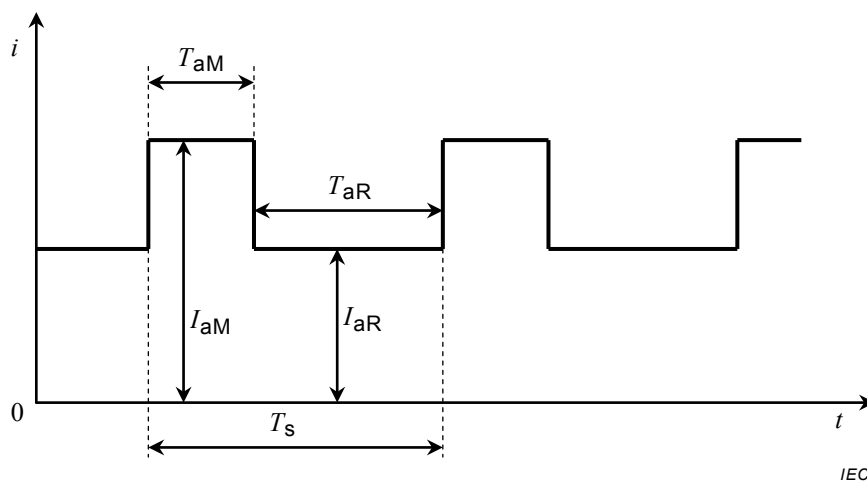


Figure 9 – Overload cycle example

For repetitive load duty, the *rated converter output current* (I_{aN1}) shall correspond, as a minimum, to the r.m.s. value of the *motor* current for a full period of the *motor* duty cycle and the *output overload capability* of the *converter* shall be adequate for the load duty cycle.

For continuous duty, the *rated converter output current* (I_{aN1}) shall correspond, as a minimum, to the continuous *motor* current necessary to supply the specified continuous *motor* torque. In the case of intermittent load duty, the overloads shall not cause the *converter* current to exceed its overload rating.

For compliance see 5.4.2.5.5

4.3.4 Operating quadrants

4.3.4.1 General

The above ratings of 4.3.2 and 4.3.3 shall be given for all *operating quadrants* (I, II, III, IV).

4.3.4.2 Operation in II and IV quadrants

The operation in II and IV quadrant applies to applications when the *motor* is running in generative mode generating power back into the *BDM/CDM/PDS* as explained in 4.2.

The input and output ratings under the operation in II and IV quadrants shall be specified including the relevant parameters for the applicable solution.

For compliance see 5.4.2.5.6

4.3.5 Ratings and properties of the control equipment

The ratings and properties of the control equipment shall be specified by the *manufacturer*.

NOTE Examples of properties: limits for voltage, current, speed and torque, *BDM/CDM* protection against motor overload and, output short circuit.

For compliance see 5.4.2.6

4.3.6 Special ratings related to *BDM/CDM/PDS* or *motor*

4.3.6.1 General

Special ratings can be specified to provide further information for specific applications or considerations. This includes the effect on the mains supply side, inside the *BDM/CDM/PDS* as well as the effect on the *motor*.

For compliance see 5.4.2.8.2, 5.4.2.8.3, 5.4.2.8.4, 5.4.2.8.5

4.3.6.2 Transformers

Power transformers may be used on the mains as well on the *motor* side of the *BDM/CDM/PDS* as step-up or step-down transformers.

The IEC 61378 series provide specifications for the design of these transformers.

IEC 61800-4 provides further information about the use of transformers in *BDM/CDM/PDS* installations.

4.4 Performance

4.4.1 Operational

4.4.1.1 General

The ratings of included features shall be specified by the *manufacturer*. One or more of the following features can be included:

- timed acceleration/ deceleration;
- *dynamic braking*;
- reversing;
- *regeneration*;
- line filtering;
- input/output data processing (analog/digital);
- automatic restart;
- *d.c. braking*.

4.4.1.2 Steady state performance

4.4.1.2.1 General

The control system is in a steady state when the reference and operating variables have been constant for more than three times the settling time of the control system and the service variables have been constant for more than the three times the longest time constant of the equipment (E.g. the thermal time constant of the speed sensor). Steady state performance for drive variables such as torque, speed, position etc. shall be specified in accordance with 4.4.1.2.2 to 4.4.1.2.6.

For compliance see 5.4.2.9

4.4.1.2.2 Deviation band

The deviation band (see Figure 10) is the total excursion of the directly controlled variable (unless another variable is specified) under steady state conditions as a result of changes in the service or operating conditions within their specified ranges.

The deviation band is expressed:

- as a percentage of the ideal maximum value of the directly controlled (or other specified) variable, see example in 4.4.1.3.3;
- as an absolute number for variables which have no readily definable base, such as position.

The signal representing the directly controlled variable should be filtered, e.g. by a first order low-pass filter with a 100 ms time constant, in order to remove noise and ripple from the signal.

NOTE The deviation band cannot be used to specify items which are not related with the steady state control performance (e.g. torque pulsation, or the speed ripple caused by load torque or *motor* torque pulsation).

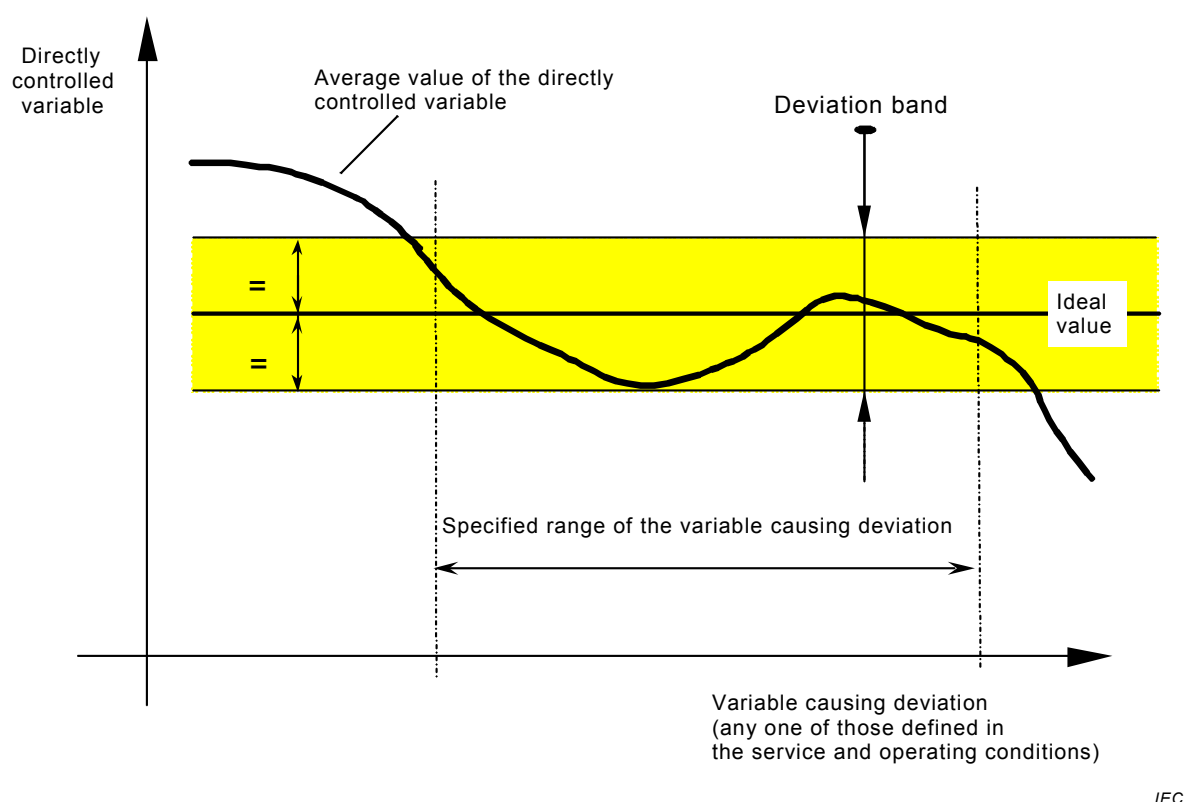


Figure 10 – Deviation band

4.4.1.2.3 Selection of deviation band

The steady state performance of a feedback control system shall be described by a number, selected from Table 8 (other levels may be defined by agreement).

The range of variables to which the deviation band applies shall be specified (see Figure 10).

Table 8 – Maximum deviation bands (percent)

±20	±10	±5	±2	±1	±0,5	±0,2	±0,1	±0,05	±0,02	±0,01
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EXAMPLE: A *PDS* has a 60 Hz – 1 780 r/min *motor* that is fed by a frequency converter. The maximum speed of the *PDS* is 2 000 r/min and the specified deviation band for the speed control is ±0,5 %. Operating conditions are, speed range: 0 to 2 000 r/min; load torque range: zero to *rated torque*. Service conditions, ambient temperature range: 5 °C to 40 °C.

Thus the deviation of the actual speed from the ideal value (speed reference) is:

$$\pm 0,5 \% \text{ of } 2\,000 \text{ r/min} = \pm 10 \text{ r/min}$$

when the value of the speed reference, load torque and ambient temperature are within their specified ranges.

E.g. if the speed reference is 1 200 r/min, the actual speed of the *motor* will be 1 200 r/min ± 10 r/min, that is between 1 190 r/min and 1 210 r/min.

4.4.1.2.4 Service deviation band – limits

The specified service deviation band (selected from Table 8) shall not be exceeded under any combination of applicable service conditions at any time during any 1 h interval following a warm-up period as specified by *the manufacturer*, with the operating variables held constant during the observation.

4.4.1.2.5 Operating deviation band – limits

The operating deviation band of the directly controlled variable (selected from Table 8) shall not be exceeded for the range of the operating variable indicated. The service conditions shall be held constant during the observation.

When required by the application, the performance information should also include data on the steady state relationship of the directly controlled variable to the reference. This aspect of performance is not included in the above discussion of operating or service deviation bands.

4.4.1.2.6 Resolution

The resolution represents the minimum obtainable variation of the controlled variable. It may be represented by an absolute value or a percentage of the maximum value.

4.4.1.3 Dynamic performance

4.4.1.3.1 General

The *BDM/CDM/PDS* shall be provided with either a current limit or a timed acceleration.

Dynamic performance shall be specified according to 4.4.1.3.2 to 4.4.1.3.3.

For compliance see 5.4.2.10.2, 5.4.2.10.3; 5.4.2.10.4, 5.4.2.10.5, 5.4.2.11 and 5.4.2.12

4.4.1.3.2 Time responses

4.4.1.3.2.1 General

Time response represents the output versus time curve resulting from the application of a specified input, under specified operating and service conditions.

The *PDS* shall operate before the application of a specified input under the following operating and service conditions:

- *maximum rated speed*;
- no load;
- *rated input voltage and input frequency*;
- temperature stabilized after a 1 h warm-up of the measuring equipment and interfaces, ambient temperature being within service conditions.

The output curve may contain a significant amount of ripple, e.g. due to the operation of the power semiconductor devices in the *BDM*. The average curve shall be used in the determination of the time response, see Figure 11. Typical time responses for a *PDS* are the time responses following a step change of speed reference, current reference or torque reference, see Figure 11, and the time response following a change in the load torque, see Figure 12. For specification purposes, the load torque of the driven equipment shall be assumed to increase linearly from zero to a specified torque (or decrease from a specified torque to zero) within 100 ms, without overshoot.

4.4.1.3.2.2 Response time

The response time is the time required, following the initiation of a specified *stimulus* to a system, for an output going in the direction of the necessary corrective action to first reach a specified value.

The specified value for a time response following a step change of reference input, see Figure 11, shall be the initial average value plus 90 % of the steady state increment. The